

Technology transfer offices play a vital role in this transition. They help protect inventions, structure licensing agreements, support spinout formation, and connect researchers with investors or industry partners. However, quantum commercialisation often stretches traditional technology transfer models because the science is complex, the development cycle is long, and the market may still be forming. Founders must therefore build teams that combine scientific depth with product management, enterprise sales, systems engineering, regulatory insight, and operational discipline. The most promising university spinouts are not merely research projects with company names; they are organisations that learn how to translate uncertain science into repeatable execution.

The so-called ‘valley of death’ between research funding and commercial traction is particularly visible in quantum. A startup may have a breakthrough patent, an excellent founding team, and early grant support, yet still struggle to prove product-market fit before its capital runs out. To survive this stage, founders often pursue phased commercialisation. They may begin by offering consulting, simulation tools, components, developer access, cloud-based experimentation, security assessment, or co-development programmes with enterprise partners. These early revenue streams may not represent the final business model, but they help the company learn from customers, refine its technology roadmap, and build credibility.

Disruptive business models and use cases

Quantum startups are experimenting with business models that reflect the practical reality of the technology. Hardware companies may follow a quantum-computing-as-a-service model, where customers access machines through the cloud rather than purchasing physical systems. This approach lowers

adoption barriers, allows centralised maintenance, and enables providers to improve hardware continuously. Other firms may sell specialised components, control systems, cryogenic equipment, photonic modules, or sensing devices to universities, laboratories, defence organisations, and industrial customers. In software, startups often build platforms that help enterprises design algorithms, simulate quantum circuits, optimise workflows, or integrate quantum experiments with classical high-performance computing.

The use cases are developing at different speeds. In pharmaceuticals and life sciences, quantum methods may eventually improve molecular simulation, protein interaction studies, and drug discovery pipelines. In finance, researchers are exploring portfolio optimisation, risk analysis, derivative pricing, and fraud detection, though these applications still require careful evidence before large-scale adoption. Logistics and manufacturing companies are interested in optimisation problems involving routing, scheduling, supply chains, and production planning. Energy and materials companies see potential in battery chemistry, catalysts, carbon capture materials, and advanced manufacturing. Quantum sensing may affect navigation, geology, medical imaging, and industrial inspection, while quantum communication and post-quantum security are becoming important as organisations prepare for future cryptographic risks.

A defining feature of the new quantum economy is that value may arrive first through hybrid workflows rather than dramatic standalone breakthroughs. A bank, pharmaceutical company, or logistics provider may not replace its classical computing environment with quantum systems. Instead, it may use quantum-inspired methods, emulators, cloud-based quantum experiments, or hybrid solvers to improve selected parts of a broader analytical process. This

gradual integration is commercially important because it creates learning loops. Enterprises become familiar with quantum concepts, startups gain domain insight, and both sides identify where quantum methods may eventually offer measurable advantage.

Intellectual property and patents in quantum

Intellectual property is becoming one of the most important strategic assets in the quantum startup economy because the sector is built on inventions that are technically complex, capital intensive, and difficult to reproduce at scale. For a quantum startup, IP is not limited to a single algorithm or device design. It may include qubit architectures, fabrication methods, control electronics, cryogenic integration, photonic packaging, ion-trap geometries, neutral-atom array control, superconducting circuit layouts, quantum networking protocols, sensing configurations, compiler methods, pulse-level optimisation techniques, and error mitigation or error correction workflows. In many cases, the commercial value of a startup depends on how clearly it can show that these inventions are differentiated, protectable, and relevant to future scalable systems.

Patents play a central role in this process, especially for hardware and enabling technologies where years of research may be required before revenue becomes substantial. A strong patent portfolio can help a startup attract venture capital, negotiate licensing agreements, form partnerships with cloud providers or semiconductor manufacturers, and defend itself against larger incumbents. Investors increasingly examine whether a quantum company has freedom to operate, whether its core claims are owned by the company or licensed from a university, and whether the patents cover practical implementation rather than only broad scientific ideas. For